

Accounting & Performance Ratio Model — Technical Specification

Caterpillar Inc. (NYSE: CAT) | FY2024

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1. Problem Statement

Caterpillar Inc. is the world's leading manufacturer of construction and mining equipment, diesel engines, and diesel-electric locomotives, operating across 190+ countries. This model analyzes CAT's FY2024 financial performance (fiscal year ended December 31, 2024) using audited data from the company's 10-K filing with the SEC. FY2023 balance sheet figures serve as prior-year comparators for denominator and average-balance calculations.

The objective is to compute and interpret 30+ ratios across five categories — performance, profitability, efficiency, leverage, and liquidity — to assess value creation, asset efficiency, and financial risk. Results feed directly into the Stage 4 AI-assisted interpretation memo.

2. Inputs (Known Variables)

Source / Item	Named Range	FY2024 (\$M)	FY2023 (\$M)
BALANCE SHEET			
Cash & cash equivalents	BAL_cash_marketable_securities_2024	6,889	—
Receivables — trade & other	BAL_receivables_2024 / _2023	9,282	9,310
Inventories	BAL_inventories_2024 / _2023	16,827	16,565
Total current assets	BAL_assets_current_2024	45,682	—
Total assets	BAL_assets_total_2024 / _2023	87,764	87,476
Total current liabilities	BAL_liabilities_current_2024	32,272	—
Total liabilities	BAL_liabilities_total_2024	68,270	—
Long-term debt (after 1 yr)	BAL_debt_long_term_2024 / _2023	27,351	24,472
Total shareholders' equity	BAL_equity_shareholders_2024 / _2023	19,494	19,503
INCOME STATEMENT			
Net sales	INC_sales	64,809	—
Cost of goods sold	INC_cost_goods_sold	43,244	—
EBIT	INC_ebit	14,187	—
Interest expense	INC_interest_expense	1,713	—
Depreciation	INC_depreciation	1,413	—
Net income			
CASH FLOW STATEMENT			
Cash from operations	CASH_operating	14,184	—
Capital expenditures	CASH_capex	(1,702)	—
Cash from financing			
MARKET / ANALYST INPUTS			
Share price (fiscal year-end)	share_price	\$349.58	—
Shares outstanding	shares_outstanding	492M	—
Cost of capital (WACC)	cost_capital	7.90%	—

Source / Item	Named Range	FY2024 (\$M)	FY2023 (\$M)
Tax rate (assumed)	tax_rate	21.40%	—

3. Named Range Conventions

Raw inputs: SOURCE_description_year (e.g., BAL_cash Marketable_securities_2024, INC_sales, CASH_operating).
 Du Pont cross-reference outputs: RATIO_asset_turnover, RATIO_operating_profit_margin, RATIO_debt_burden, RATIO_leverage. Derived intermediates: startYear_, currentYear_, and avg_ prefixes. All named ranges are defined in the Excel Name Manager; no bare cell references appear in the Ratios tab.

4. Assumptions & Constraints

- Tax rate assumed at statutory 21.4%; not derived from INC_taxes / INC_taxable_income.
- Interest expense on simple annual basis; no sub-period compounding.
- No off-balance-sheet items included; operating leases and pension obligations not capitalized.
- Total capitalization = long-term debt (after 1 yr) + shareholders' equity — excludes short-term debt (\$4.4B) and current LTD (\$6.7B).
- Quick ratio uses trade receivables only; Cat Financial finance receivables (\$9.6B) excluded.
- WACC of 7.9% is an analyst estimate with no formal CAPM build-up.
- Share price (\$349.58) reflects fiscal year-end close; market cap = share_price × shares_outstanding, no diluted adjustment.
- Average balances = simple average of FY2023 and FY2024 period-end figures.

5. Calculation Flow

Derived Inputs

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market_cap = share_price × shares_outstanding
startYear_equity = BAL_equity_shareholders_2023
startYear_total_assets = BAL_assets_total_2023
startYear_total_cap = BAL_debt_long_term_2023 + BAL_equity_shareholders_2023
startYear_inventory = BAL_inventories_2023
startYear_receivables = BAL_receivables_2023
currentYear_after_tax_op_inc = INC_net + (1 - tax_rate) × INC_interest_expense
currentYear_daily_sales = INC_sales / 365
currentYear_daily_cogs = INC_cost_goods_sold / 365
currentYear_nwc = BAL_assets_current_2024 - BAL_liabilities_current_2024
currentYear_total_cap = BAL_debt_long_term_2024 + BAL_equity_shareholders_2024
avg_equity = AVERAGE(startYear_equity, BAL_equity_shareholders_2024)
avg_total_assets = AVERAGE(startYear_total_assets, BAL_assets_total_2024)
avg_total_cap = AVERAGE(startYear_total_cap, currentYear_total_cap)

```

Performance

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MVA = market_cap - BAL_equity_shareholders_2024
M/B = market_cap / BAL_equity_shareholders_2024
EVA = currentYear_after_tax_op_inc - (cost_capital × startYear_total_cap)

```

Profitability (start-of-year and average-based)

$ROA = \text{currentYear_after_tax_op_inc} / \text{startYear_total_assets}$ [avg: / avg_total_assets]
 $ROC = \text{currentYear_after_tax_op_inc} / \text{startYear_total_cap}$ [avg: / avg_total_cap]
 $ROE = \text{INC_net} / \text{startYear_equity}$ [avg: / avg_equity]

Efficiency

$RATIO_asset_turnover = \text{INC_sales} / \text{startYear_total_assets}$
 $\text{Receivables turnover} = \text{INC_sales} / \text{startYear_receivables}$
 $\text{Avg collection period (days)} = \text{startYear_receivables} / \text{currentYear_daily_sales}$
 $\text{Inventory turnover} = \text{INC_cost_goods_sold} / \text{startYear_inventory}$
 $\text{Days in inventory} = \text{startYear_inventory} / \text{currentYear_daily_cogs}$
 $\text{Profit margin} = \text{INC_net} / \text{INC_sales}$
 $RATIO_operating_profit_margin = \text{currentYear_after_tax_op_inc} / \text{INC_sales}$

Leverage

$LT \text{ debt ratio} = \text{BAL_debt_long_term_2024} / \text{currentYear_total_cap}$
 $LT \text{ debt-equity ratio} = \text{BAL_debt_long_term_2024} / \text{BAL_equity_shareholders_2024}$
 $\text{Total debt ratio} = \text{BAL_liabilities_total_2024} / \text{BAL_assets_total_2024}$
 $\text{Times interest earned} = \text{INC_ebit} / \text{INC_interest_expense}$
 $\text{Cash coverage ratio} = (\text{INC_ebit} + \text{INC_depreciation}) / \text{INC_interest_expense}$
 $RATIO_debt_burden = \text{INC_net} / \text{currentYear_after_tax_op_inc}$
 $RATIO_leverage = \text{BAL_assets_total_2024} / \text{BAL_equity_shareholders_2024}$

Liquidity

$NWC \text{ to assets} = \text{currentYear_nwc} / \text{BAL_assets_total_2024}$
 $\text{Current ratio} = \text{BAL_assets_current_2024} / \text{BAL_liabilities_current_2024}$
 $\text{Quick ratio} = (\text{BAL_cash_marketable_securities_2024} + \text{BAL_receivables_2024}) / \text{BAL_liabilities_current_2024}$
 $\text{Cash ratio} = \text{BAL_cash_marketable_securities_2024} / \text{BAL_liabilities_current_2024}$

Du Pont

$ROA \text{ (Du Pont)} = RATIO_asset_turnover \times RATIO_operating_profit_margin$
 $ROE \text{ (Du Pont)} = RATIO_leverage \times RATIO_asset_turnover \times RATIO_operating_profit_margin \times RATIO_debt_burden$

6. Outputs

- Scalar inputs table: All derived intermediates and averages labeled with named ranges and values.
- Ratio output table: 30 ratios in six categories with named-range formula references and numeric results.
- Du Pont tree: Decomposition confirming ROA and ROE via RATIO_* cross-references to source sections.
- Balance check: Total Assets = Total Liabilities + Equity verified for FY2024 (\$87,764M) and FY2023 (\$87,476M).
- Notes tab: Assumptions log, data source citation, color-coding legend, and full record of Stage 2 formula corrections.

7. Model Review — What Worked & What to Improve

What was built correctly

The named-range architecture is the model's strongest feature — every input is descriptively named, the Du Pont section cross-references `RATIO_*` ranges rather than bare cell addresses, and the Notes tab logs every formula correction with a plain-language explanation. Balance sheet checks pass for both years. Start-of-year vs. average-based profitability variants are consistently maintained.

What to improve

- Quick ratio excludes Cat Financial finance receivables (\$9.6B) without an inline cell note — embed documentation directly.
- Total capitalization omits short-term debt (\$4.4B) and current LTD (\$6.7B); label the metric explicitly or include all debt.
- Tax rate should be derived as `INC_taxes / INC_taxable_income` rather than manually assumed at 21.4%.
- WACC has no build-up; add a sensitivity table (6%–10%) to stress-test EVA for a CFO audience.
- Layout: freeze panes at category headers, add conditional formatting for out-of-norm ratios, group rows by category.
- Add free cash flow (`CASH_operating – CASH_capex = $12,482M`) to contextualize the \$7.8B share repurchase.

8. Limitations & Next Steps

Excluded: Multi-year trend analysis, peer benchmarking (Deere & Co., CNH Industrial), Cat Financial segment-level adjustments, off-balance-sheet capitalization (operating leases, pension), and scenario/sensitivity analysis.

Stage 4 setup: Section 5's Calculation Flow becomes the AI prompt instruction block. Section 7's Model Review directs the AI to flag the four assumptions most requiring commentary: WACC sensitivity, capitalization scope, quick ratio composition, and tax rate derivation. Section 6's Outputs define the ratio table the AI will interpret to produce a CFO- and investor-facing narrative on CAT's value creation, operational efficiency, capital structure risk, and liquidity position.

Data source: Caterpillar Inc., 10-K FY2024, SEC EDGAR. All figures in millions of USD unless noted. Model built in Microsoft Excel with named-range conventions.